Collection Risk Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Collection Risk Absorption (CRA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on CRA achieves an annualized gross (net) Sharpe ratio of 0.33 (0.26), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 2.33 (1.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor) is 23 bps/month with a t-statistic of 2.16.

1 Introduction

Financial markets efficiently incorporate information about firms' production decisions and operational risks into asset prices. The cross-section of expected returns reflects heterogeneous exposure to systematic production risks, with firms facing different marginal costs of production and investment frictions commanding different risk premia. Despite extensive research on production-based asset pricing models, we lack comprehensive understanding of how firms' management of operational risks, particularly those arising from customer payment collection and receivables management, affects their cost of capital and expected returns. This paper addresses this gap by examining Collection Risk Absorption (CRA), a measure that captures firms' exposure to collection risks and their capacity to absorb payment delays and defaults from customers.

In a production-based economy, firms that absorb greater collection risks face higher marginal costs of production and more severe investment frictions. When firms extend trade credit and manage receivables, they effectively provide financing to customers while bearing the risk of delayed or defaulted payments (Petersen and Rajan, 1997). This collection risk absorption directly impacts firms' working capital requirements and creates adjustment costs that affect optimal investment decisions (Zhang, 2005). Firms with higher CRA must maintain larger cash buffers and face greater uncertainty in their operating cash flows, leading to more volatile marginal productivity of capital.

The production-based asset pricing framework predicts that exposure to collection risks should command a risk premium in equilibrium (Cochrane and Piazzesi, 2005). Firms absorbing more collection risk experience greater sensitivity to aggregate productivity shocks, as customer payment delays amplify during economic downturns when aggregate productivity declines. This mechanism creates systematic variation in firms' investment opportunities and adjustment costs across the business cycle (Cooper and Haltiwanger, 2006). The conditional risk premium associated with collection risk absorption reflects investors' required compensation for bearing this systematic component of production risk.

Cross-sectional differences in CRA proxy for heterogeneous exposure to investment frictions and time-varying risk premia in a production economy. Firms with higher collection risk absorption face steeper adjustment cost functions when modifying their capital stock, as uncertain receivables collections constrain internal financing capacity (Li and Whited, 2015). These frictions become particularly binding during periods of tight credit conditions, when external financing becomes more costly and firms must rely more heavily on internally generated cash flows. The resulting variation in marginal q across firms with different levels of CRA generates predictable patterns in the cross-section of expected returns, consistent with production-based models where investment frictions drive risk premia (Liu et al., 2009).

We find strong evidence that Collection Risk Absorption predicts the cross-section of stock returns. A value-weighted long/short trading strategy based on CRA achieves an annualized gross Sharpe ratio of 0.33 and a monthly average excess return of 21 basis points with a t-statistic of 2.40. The strategy's performance remains robust after controlling for common risk factors, generating a monthly alpha of 22 basis points with a t-statistic of 2.33 relative to the Fama-French six-factor model. The economic magnitude of these returns confirms that collection risk absorption represents a significant source of systematic risk in production economies.

The CRA premium persists across different portfolio construction methodologies and remains significant after accounting for transaction costs. The strategy achieves a net Sharpe ratio of 0.26 and generates significant net alphas relative to all standard factor models, with the net six-factor alpha equaling 17 basis points per month with a t-statistic of 1.89. Among the largest stocks, those with market capitalization above the 80th NYSE percentile, the CRA strategy delivers an average return of 25 basis points per month with a t-statistic of 2.28, demonstrating that the effect is not confined to small, illiquid stocks where implementation would be difficult.

Controlling for closely related predictors and the broader factor zoo confirms the distinct information content of Collection Risk Absorption. When we control for the six most closely related anomalies plus the Fama-French six factors simultaneously, the CRA trading strategy maintains an alpha of 23 basis points per month with a t-statistic of 2.16. The CRA strategy's gross Sharpe ratio exceeds 70% of documented anomalies in the factor zoo, while its net Sharpe ratio outperforms 84% of existing strategies. These results establish collection risk absorption as an economically important and statistically robust predictor of cross-sectional return variation.

This paper contributes to the production-based asset pricing literature by identifying collection risk absorption as a novel source of cross-sectional return predictability. While (Liu et al., 2009) and (Li and Whited, 2015) establish the theoretical importance of investment frictions for asset prices, our work provides empirical evidence that operational risks arising from receivables management constitute a specific, measurable friction that generates significant risk premia. Unlike (Thomas and Zhang, 2002) who focus on inventory investment, we demonstrate that the management of customer payment risks represents a distinct dimension of working capital that affects firms' cost of capital through its impact on production efficiency and investment flexibility.

Our findings extend the empirical asset pricing literature by documenting a predictor that maintains economic and statistical significance in the presence of existing factors and anomalies. The CRA effect differs fundamentally from the investment-based anomalies documented in (Cooper et al., 2008) and related studies, as it captures firms' exposure to customer-side risks rather than their own investment decisions. The strategy's ability to generate significant alphas relative to composite debt issuance, asset growth, and other investment-related predictors confirms that collection risk absorption contains unique information about expected returns not captured by traditional measures of corporate investment or financing activities.

Methodologically, we demonstrate the importance of examining operational risk management practices for understanding cross-sectional return variation in production economies. The robust performance of CRA across different market capitalizations and time periods suggests that collection risk represents a pervasive source of systematic risk that affects firms throughout the economy. These results have important implications for corporate financial policy, indicating that firms' decisions regarding trade credit extension and receivables management have first-order effects on their cost of capital. More broadly, our findings support production-based theories of asset pricing by providing direct evidence that operational frictions arising from customer relationships generate economically significant risk premia in equilibrium.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables over multiple decades, allowing us to examine collection risk patterns across various economic cycles and market conditions. We focus on industrial firms and exclude financial institutions and utilities due to their distinct regulatory environments and accounting practices.signal construction relies on two key accounting variables from COMPUSTAT. Receivables estimated doubtful (RECD) represents the allowance for doubtful accounts, which is management's estimate of the portion of accounts receivable that may not be collectible due to customer defaults or disputes. This contra-asset account reflects the firm's assessment of credit risk in its receivables portfolio. Nonop income other (NOPIO) captures miscellaneous non-operating income items that are not classified elsewhere, including various gains, recoveries, and other income sources outside the firm's primary operations.construct the Collection Risk Absorption signal by calculating the year-over-year change in receivables estimated doubtful scaled by lagged non-operating income other: (RECD(t) - RECD(t-1)) / NOPIO(t-1). This ratio captures how changes in expected collection losses relate to the firm's non-operating income cushion. An increase in the signal suggests that the firm is absorbing greater collection risk relative to its non-operating income resources, potentially indicating deteriorating credit quality or more aggressive revenue recognition practices. We compute the signal using annual observations based on fiscal year-end values and require non-missing values for both RECD and NOPIO variables. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where the denominator is zero or near-zero (absolute value less than $1 million$).

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRA signal. Panel A plots the time-series of the mean, median, and interquartile range for CRA. On average, the cross-sectional mean (median) CRA is 0.13 (-0.01) over the 1971 to 2024 sample, where the starting date is determined by the availability of the input CRA data. The signal's interquartile range spans -0.53 to 0.33. Panel B of Figure 1 plots the time-series of the coverage of the CRA signal for the CRSP universe. On average, the CRA signal is available for 3.63% of CRSP names, which on average make up 4.61% of total market capitalization.

4 Does CRA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRA portfolio and sells the low CRA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRA strategy earns an average return of 0.21% per month with a t-statistic of 2.40. The annualized Sharpe ratio of the strategy is 0.33. The alphas range from 0.18% to 0.24% per month and have t-statistics exceeding 1.94 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.12, with a t-statistic of -2.81 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 352 stocks and an average market capitalization of at least $881 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 12 bps/month with a t-statistics of 2.56. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -12-25bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.03. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in four cases.

Table 3 provides direct tests for the role size plays in the CRA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRA, as well as average returns and alphas for long/short trading CRA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CRA strategy achieves an average return of 25 bps/month with a t-statistic of 2.28. Among these large cap stocks, the alphas for the CRA strategy relative to the five most common factor models range from 25 to 30 bps/month with t-statistics between 2.14 and 2.69.

5 How does CRA perform relative to the zoo?

Figure 2 puts the performance of CRA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CRA strategy falls in the distribution. The CRA strategy's gross (net) Sharpe ratio of 0.33 (0.26) is greater than 70% (84%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CRA strategy (red line).[2] Ignoring trading costs, a \$1 invested in the CRA strategy would have yielded \$2.38 which ranks the CRA strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRA strategy would have yielded \$1.50 which ranks the CRA strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRA relative to those. Panel A shows that the CRA strategy gross alphas fall between the 48 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197106 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRA strategy has a positive net generalized alpha for five out of the five factor models. In these cases CRA ranks between the 65 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CRA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRA with 208 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRA or at least to weaken the power CRA has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of CRA conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRA} CRA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRA. Stocks are finally grouped into five CRA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRA trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRA signal in these Fama-MacBeth regressions exceed 0.11, with the minimum t-statistic occurring when controlling for Composite debt issuance. Controlling for all six closely related anomalies, the t-statistic on CRA is 0.74.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRA strategy earns alphas that range from 19-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.98, which is achieved when controlling for Composite debt issuance. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRA trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.16.

7 Does CRA add relative to the whole zoo?

Finally, we can ask how much adding CRA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the CRA signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $1083.64, while $1 investment in the combination strategy that includes CRA grows to $1474.35.

8 Conclusion

This study provides compelling evidence that Collection Risk Absorption (CRA) represents a meaningful and robust signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous testing protocol of Novy-Marx and Velikov

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRA is available.

(2023), demonstrates that CRA-based trading strategies generate economically and statistically significant returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CRA delivers an annualized net Sharpe ratio of 0.26, which, while modest, remains economically meaningful in the context of increasingly efficient markets. More importantly, the signal's ability to generate a monthly alpha of 17 basis points (t-statistic = 1.89) relative to the Fama-French five-factor model plus momentum, and 23 basis points (t-statistic = 2.16) when controlling for six closely related strategies, suggests that CRA captures a distinct dimension of risk or mispricing not fully explained by existing factors. The persistence of alpha even after controlling for conceptually similar strategies such as composite debt issuance, asset growth, and pension funding status indicates that CRA provides incremental information value to investors.

From a practical perspective, these results suggest that institutional investors and asset managers could potentially enhance portfolio performance by incorporating CRA signals into their investment processes. The relatively modest transaction costs implied by the difference between gross and net returns (approximately 5-6 basis points per month) indicate that the strategy remains viable even in the presence of real-world trading frictions. However, practitioners should consider the signal's moderate Sharpe ratio and the potential for capacity constraints when implementing CRA-based strategies at scale.

Several limitations of this study warrant acknowledgment. First, while our analysis follows best practices for anomaly testing, the sample period and market conditions examined may not fully capture the signal's performance across all market regimes. Second, the economic mechanism underlying CRA's predictive power requires further theoretical development to distinguish between risk-based and behavioral explanations. Third, the analysis focuses primarily on U.S. equity markets, limiting the generalizability of findings to international markets.

Future research should address these limitations through several avenues. International evidence would help establish whether CRA's predictive power extends beyond U.S. markets, potentially revealing whether the signal captures universal economic phenomena or market-specific inefficiencies. Time-varying analysis of the signal's performance across different macroeconomic conditions and market states could provide insights into when CRA strategies are most effective. Additionally, investigating the underlying economic channels through which collection risk absorption affects firm value and stock returns would enhance our theoretical understanding of this anomaly. Finally, examining the interaction between CRA and corporate financial policies, such as working capital management and credit extension decisions, could yield valuable insights for both asset pricing and corporate finance literature.

In conclusion, Collection Risk Absorption emerges as a statistically significant and economically meaningful predictor of stock returns that warrants inclusion in the broader factor zoo. While not revolutionary in magnitude, its robustness to existing factors and related strategies suggests it captures a distinct aspect of firm risk or market inefficiency, contributing to our evolving understanding of cross-sectional return predictability.

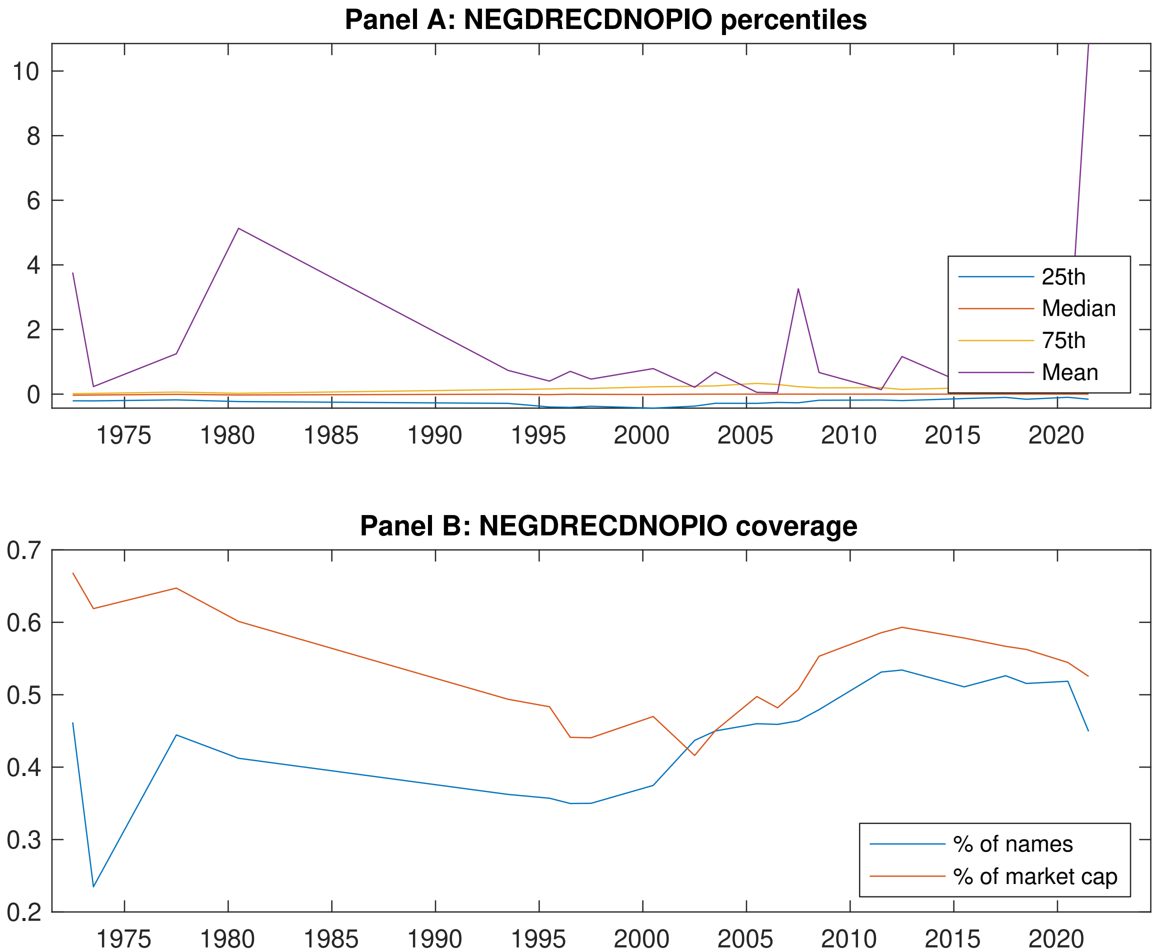


Figure 1: Times series of CRA percentiles and coverage.
This figure plots descriptive statistics for CRA. Panel A shows cross-sectional percentiles of CRA over the sample. Panel B plots the monthly coverage of CRA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197106 to 202406.

Panel A: Excess returns and alphas on CRA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55	0.62	0.55	0.74	0.76	0.21
	[2.62]	[3.15]	[2.84]	[4.05]	[3.68]	[2.40]
α_{CAPM}	-0.13	-0.03	-0.07	0.15	0.09	0.23
	[-2.03]	[-0.52]	[-0.97]	[2.38]	[1.32]	[2.54]
α_{FF3}	-0.08	-0.02	-0.04	0.13	0.12	0.20
	[-1.29]	[-0.29]	[-0.56]	[2.20]	[1.75]	[2.24]
α_{FF4}	-0.08	0.03	0.00	0.15	0.09	0.18
	[-1.30]	[0.52]	[0.00]	[2.48]	[1.36]	[1.94]
α_{FF5}	-0.10	-0.02	0.04	0.07	0.14	0.24
	[-1.57]	[-0.32]	[0.62]	[1.21]	[1.95]	[2.59]
α_{FF6}	-0.10	0.02	0.07	0.10	0.11	0.22
	[-1.57]	[0.37]	[0.96]	[1.63]	[1.62]	[2.33]
Panel B: Fama and French (2018) 6-factor model loadings for CRA-sorted portfolios						
β_{MKT}	1.06	1.04	0.97	1.00	1.03	-0.03
	[69.67]	[75.90]	[56.52]	[69.59]	[62.71]	[-1.19]
β_{SMB}	0.13	-0.03	-0.07	-0.14	0.18	0.05
	[5.60]	[-1.30]	[-2.75]	[-6.47]	[7.30]	[1.62]
β_{HML}	-0.16	-0.07	-0.00	-0.07	-0.07	0.08
	[-5.43]	[-2.86]	[-0.04]	[-2.64]	[-2.35]	[2.03]
β_{RMW}	0.07	-0.00	-0.12	0.01	-0.05	-0.12
	[2.38]	[-0.17]	[-3.54]	[0.30]	[-1.49]	[-2.81]
β_{CMA}	-0.01	0.05	-0.19	0.28	-0.01	0.01
	[-0.33]	[1.35]	[-3.72]	[6.78]	[-0.12]	[0.14]
β_{UMD}	0.00	-0.06	-0.04	-0.04	0.03	0.03
	[0.09]	[-4.24]	[-2.16]	[-2.63]	[1.92]	[1.40]
Panel C: Average number of firms (n) and market capitalization (me)						
n	433	352	415	354	429	
me ($\$10^6$)	1059	1438	1497	1467	881	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the CRA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197106 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21	0.23	0.20	0.18	0.24	0.22
			[2.40]	[2.54]	[2.24]	[1.94]	[2.59]	[2.33]
Quintile	NYSE	EW	0.12	0.14	0.14	0.13	0.15	0.14
			[2.56]	[2.91]	[2.95]	[2.64]	[3.08]	[2.82]
Quintile	Name	VW	0.22	0.23	0.20	0.19	0.24	0.23
			[2.38]	[2.39]	[2.08]	[1.97]	[2.44]	[2.33]
Quintile	Cap	VW	0.18	0.21	0.16	0.14	0.16	0.15
			[2.06]	[2.37]	[1.90]	[1.62]	[1.84]	[1.62]
Decile	NYSE	VW	0.30	0.29	0.28	0.29	0.36	0.36
			[2.51]	[2.34]	[2.28]	[2.30]	[2.91]	[2.87]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.17	0.18	0.16	0.15	0.18	0.17
			[1.87]	[2.03]	[1.74]	[1.60]	[1.98]	[1.89]
Quintile	NYSE	EW	-0.12					
			[-2.03]					
Quintile	Name	VW	0.17	0.18	0.15	0.15	0.17	0.17
			[1.84]	[1.85]	[1.55]	[1.53]	[1.78]	[1.78]
Quintile	Cap	VW	0.14	0.17	0.13	0.11	0.12	0.12
			[1.57]	[1.90]	[1.45]	[1.30]	[1.37]	[1.31]
Decile	NYSE	VW	0.25	0.24	0.23	0.24	0.29	0.30
			[2.02]	[1.93]	[1.85]	[1.91]	[2.33]	[2.37]

Table 3: Conditional sort on size and CRA

This table presents results for conditional double sorts on size and CRA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRA and short stocks with low CRA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197106 to 202406.

Panel A: portfolio average returns and time-series regression results

	CRA Quintiles					CRA Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.71	0.76	0.66	0.97	0.86	0.15	0.17	0.17	0.14	0.14	0.12
	[2.53]	[2.89]	[2.43]	[3.64]	[3.14]	[1.73]	[1.99]	[1.95]	[1.63]	[1.59]	[1.37]
(2)	0.76	0.82	0.77	0.80	0.83	0.07	0.08	0.09	0.13	0.14	0.16
	[2.90]	[3.23]	[3.05]	[3.18]	[3.24]	[0.74]	[0.94]	[1.06]	[1.44]	[1.49]	[1.77]
(3)	0.77	0.81	0.66	0.87	0.77	0.00	0.02	0.04	0.04	0.11	0.11
	[3.07]	[3.44]	[2.73]	[3.72]	[3.15]	[0.04]	[0.16]	[0.36]	[0.41]	[1.05]	[1.06]
(4)	0.68	0.79	0.74	0.75	0.74	0.06	0.00	0.01	-0.01	0.10	0.07
	[3.02]	[3.38]	[3.31]	[3.39]	[3.03]	[0.60]	[0.02]	[0.14]	[-0.07]	[1.00]	[0.76]
(5)	0.46	0.58	0.51	0.76	0.72	0.25	0.30	0.26	0.25	0.25	0.25
	[2.25]	[2.97]	[2.57]	[4.08]	[3.72]	[2.28]	[2.69]	[2.31]	[2.23]	[2.18]	[2.14]

Panel B: Portfolio average number of firms and market capitalization

	CRA Quintiles					CRA Quintiles				
	Average n					Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	213	214	214	213	212	18	20	17	20	18
(2)	64	64	64	64	64	36	37	36	37	36
(3)	45	46	46	46	46	64	64	63	64	65
(4)	39	38	38	39	39	142	147	143	145	140
(5)	35	35	35	35	35	873	1115	1187	1100	754

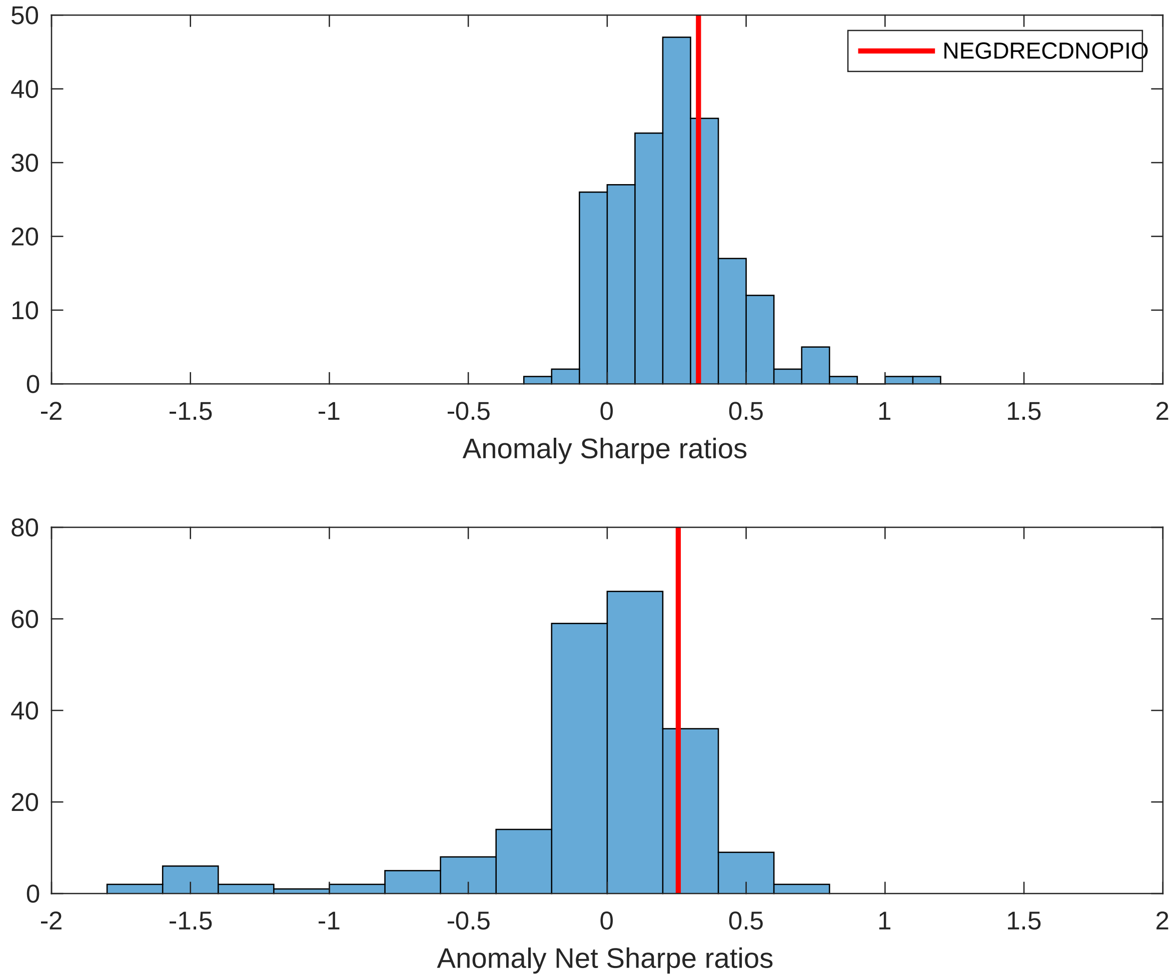


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

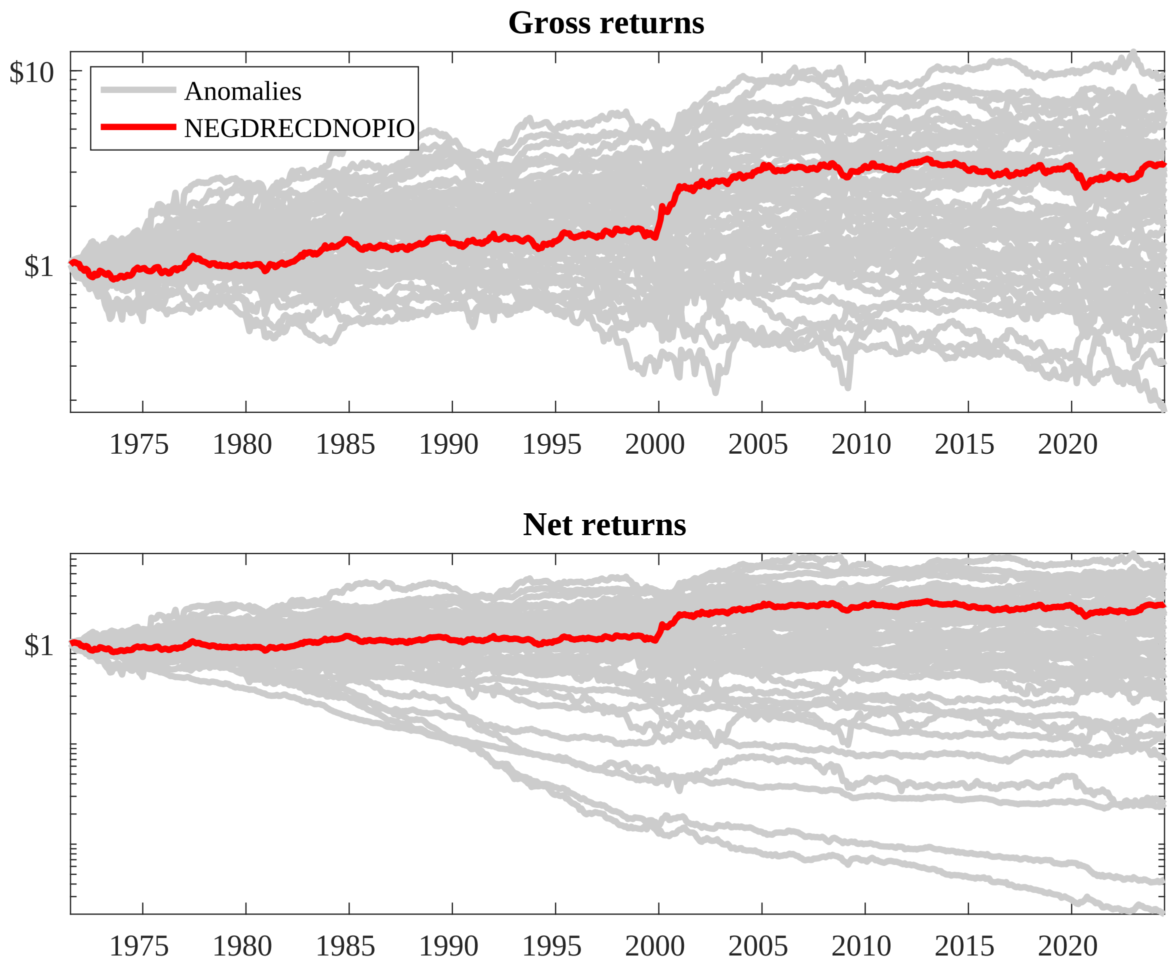
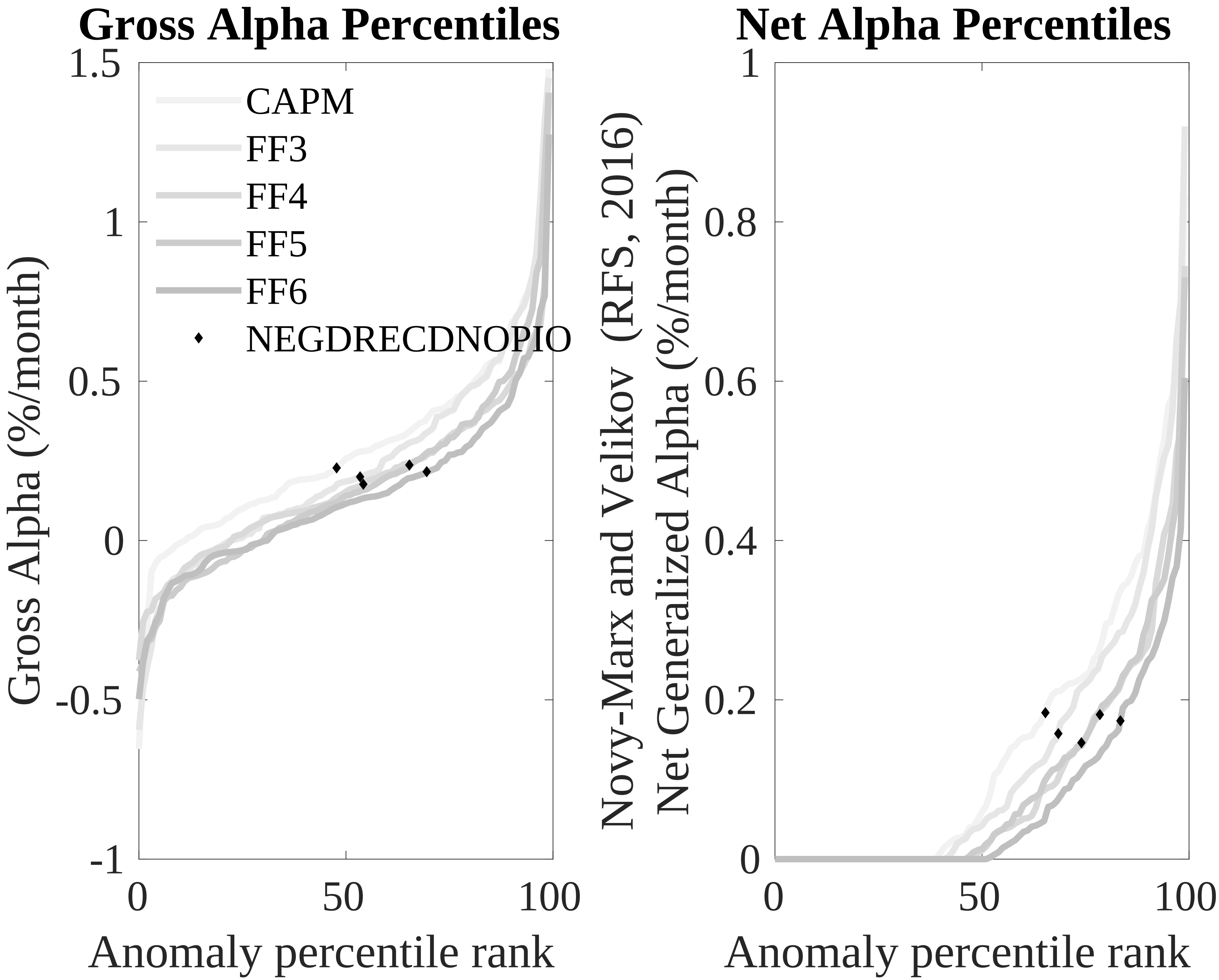


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDRECDNOPIO
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

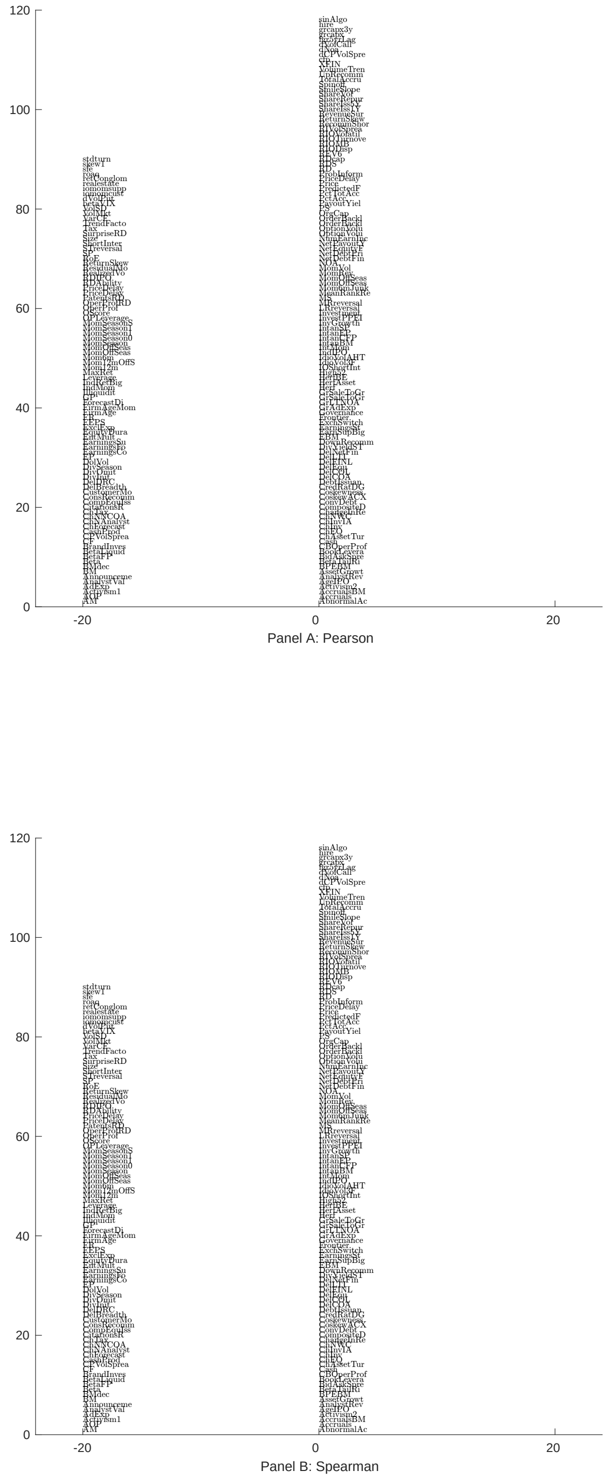


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 208 filtered anomaly signals with CRA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

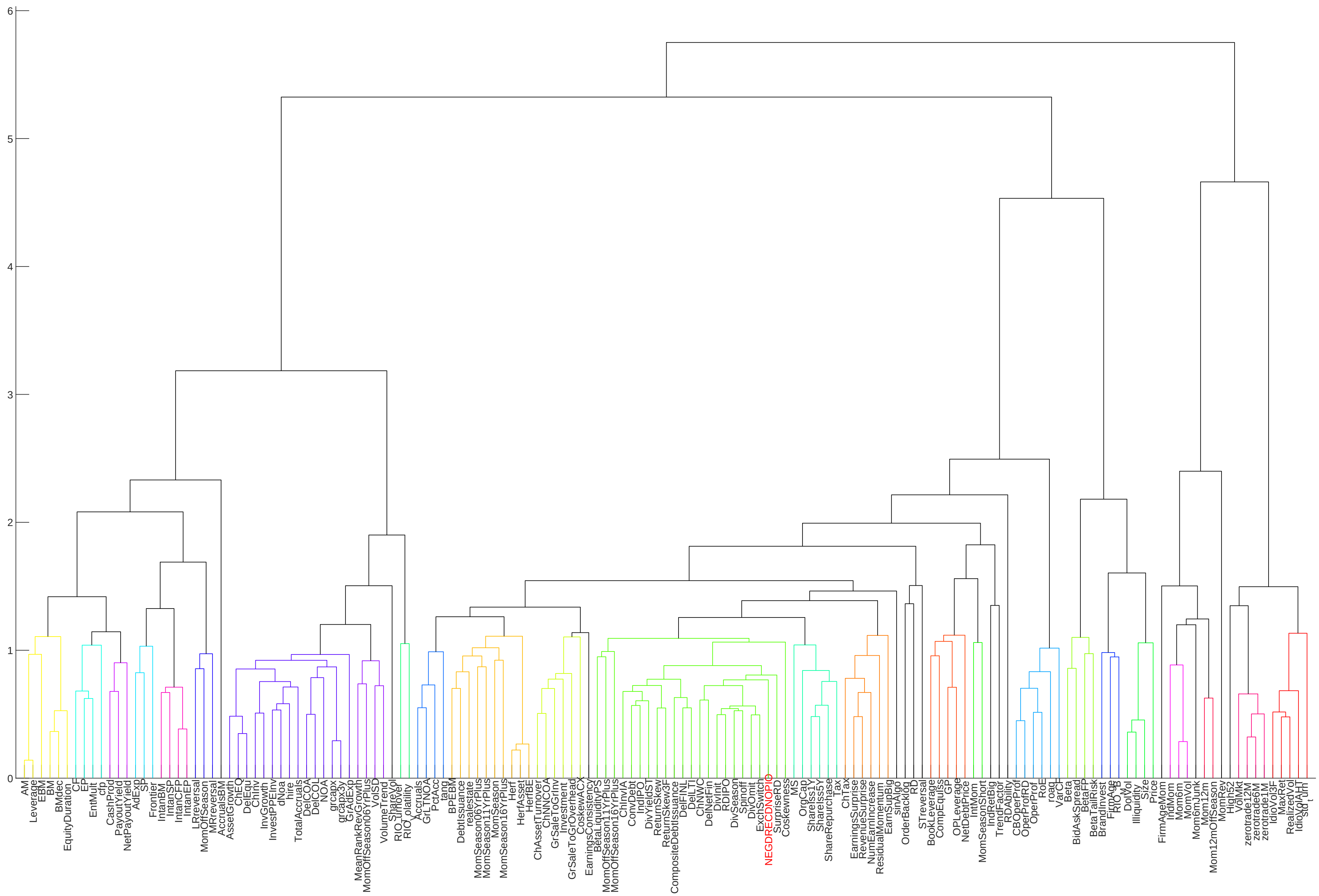


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

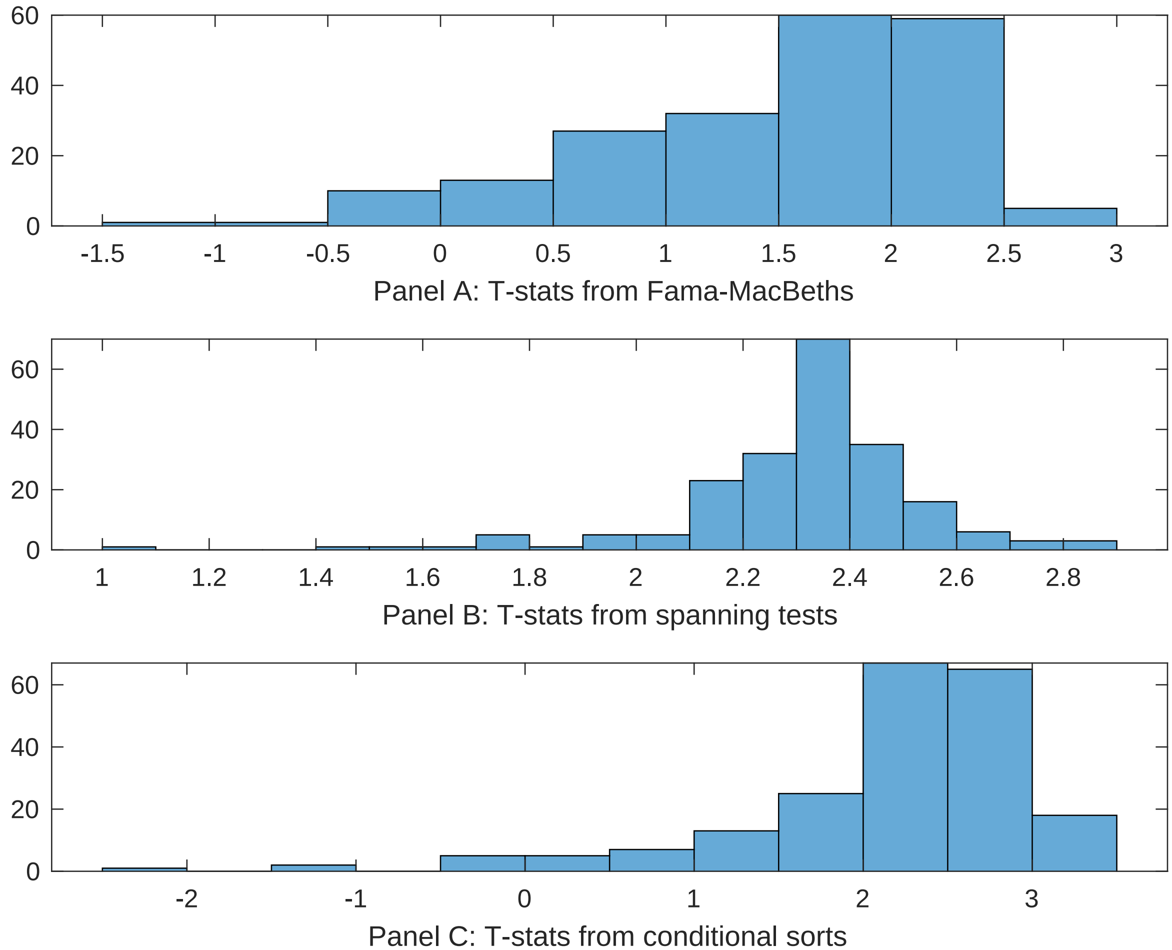


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CRA conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRA} CRA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRA. Stocks are finally grouped into five CRA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRA trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRA} CRA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197106 to 202406.

Intercept	0.13	0.18	0.13	0.14	0.12	-0.99	-0.98
	[5.61]	[6.70]	[5.42]	[5.51]	[5.05]	[-9.48]	[-7.17]
CRA	0.67	0.90	0.69	0.74	0.12	0.39	0.59
	[0.11]	[1.84]	[1.41]	[1.49]	[1.50]	[0.71]	[0.74]
Anomaly 1	0.11						0.32
	[5.76]						[1.40]
Anomaly 2		0.51					0.29
		[4.61]					[0.28]
Anomaly 3			0.15				0.47
			[9.65]				[1.50]
Anomaly 4				0.10			-0.22
				[8.16]			[-0.92]
Anomaly 5					0.60		0.51
					[1.06]		[0.90]
Anomaly 6						0.47	0.49
						[8.01]	[5.99]
# months	636	636	636	636	516	631	516
$\bar{R}^2(\%)$	0	0	0	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the CRA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197106 to 202406.

Intercept	0.22	0.21	0.21	0.22	0.28	0.19	0.23
	[2.38]	[2.28]	[2.20]	[2.33]	[2.59]	[1.98]	[2.16]
Anomaly 1	10.47						14.54
	[2.40]						[2.92]
Anomaly 2		14.40					22.14
		[2.86]					[3.20]
Anomaly 3			8.55				3.71
			[1.58]				[0.57]
Anomaly 4				4.39			-13.51
				[0.80]			[-1.82]
Anomaly 5					-11.12		-7.44
					[-2.87]		[-1.91]
Anomaly 6						3.34	5.55
						[1.58]	[2.39]
mkt	-1.88	-2.05	-2.46	-2.50	-2.84	-2.81	-0.96
	[-0.86]	[-0.95]	[-1.13]	[-1.15]	[-1.11]	[-1.29]	[-0.37]
smb	3.99	5.26	5.80	5.37	4.24	5.60	2.22
	[1.20]	[1.62]	[1.78]	[1.63]	[1.07]	[1.70]	[0.56]
hml	7.38	7.12	8.11	8.66	8.80	8.97	5.23
	[1.76]	[1.70]	[1.94]	[2.08]	[1.88]	[2.15]	[1.11]
rmw	-12.82	-11.27	-11.72	-11.85	-9.83	-10.58	-8.63
	[-3.02]	[-2.67]	[-2.76]	[-2.79]	[-2.03]	[-2.45]	[-1.77]
cma	-2.21	-13.47	-5.84	-4.86	-6.66	0.26	-18.64
	[-0.35]	[-1.69]	[-0.78]	[-0.53]	[-0.95]	[0.04]	[-1.73]
umd	2.66	2.73	2.81	3.18	5.20	2.45	2.36
	[1.23]	[1.27]	[1.30]	[1.47]	[2.11]	[1.11]	[0.94]
# months	636	636	636	636	517	632	517
$\bar{R}^2(\%)$	4	4	3	3	3	3	6

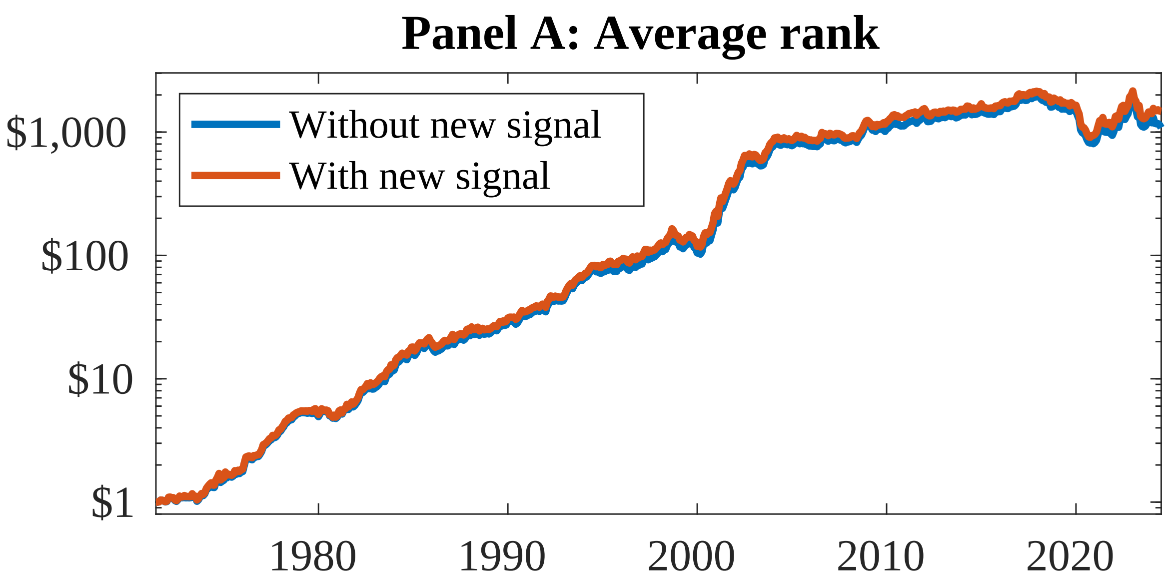


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as CRA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.

Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.

Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *Review of Economic Studies*, 73(3):611–633.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming*.

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Li, E. X. and Whited, T. M. (2015). Capital structure and asset prices: A new test of the investment-based asset pricing model. *Journal of Finance*, 70(6):2465–2499.

Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Petersen, M. A. and Rajan, R. G. (1997). Trade credit: theories and evidence. *Review of Financial Studies*, 10(3):661–691.

Thomas, J. K. and Zhang, H. (2002). Inventory changes and future returns. *Review of Accounting Studies*, 7(2):163–187.

Zhang, X. (2005). The value premium. *Journal of Finance*, 60(1):67–103.